

In 1986, the Voyager 2 space probe took these close-ups of Uranus's five largest moons, including Ariel and Umbriel, discovered in 1851, and the smallest, called Miranda, which was discovered in 1948.

IN 1849, FRENCH ASTRONOMER EDOUARD ROCHE (1820–83)

explained why Saturn has rings as well as moons. If planets and moons get too close to Saturn they are **ripped apart by tidal forces**—the changing pull of gravity created as they all rotate. There is a limit to how close planets and moons can get without being pulled apart; this became known as the **Roche limit**. If the planets and moons have identical densities, the Roche limit is 2.446 times the radius of the planet. If our Moon ever strayed closer to Earth than about 11,476 miles (18,470 km), it too would be shredded into rings.

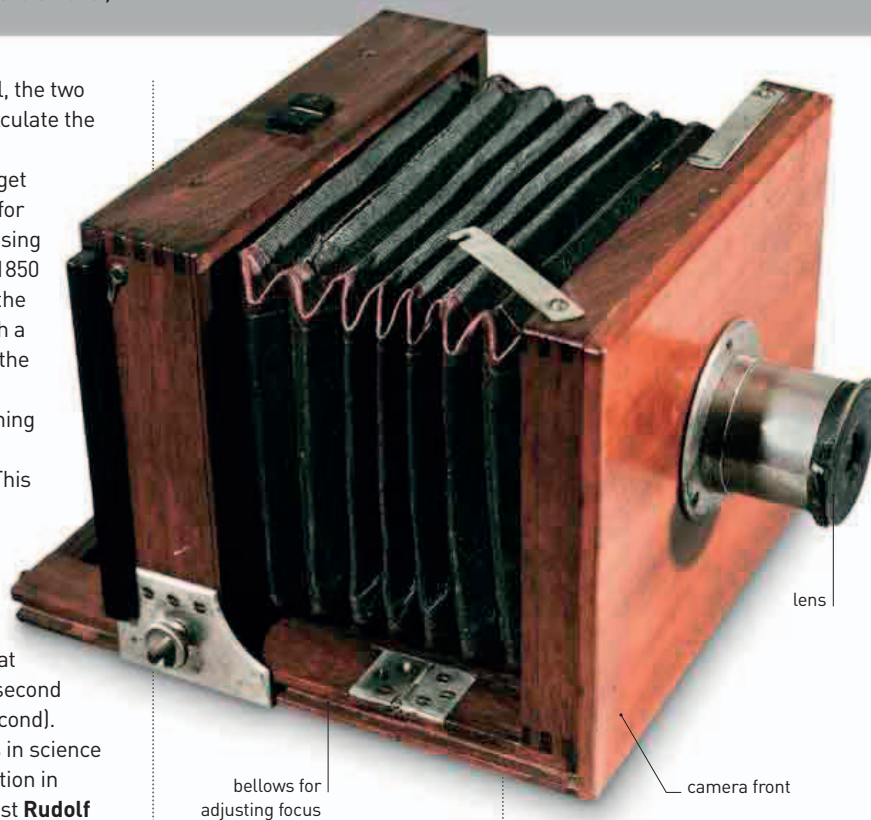
Two other French scientists, physicists **Hippolyte Fizeau** (1819–96) and **Jean Foucault** (1819–68), **measured the speed of light** in 1849 by bouncing a beam of light off a mirror, through slots on a rapidly rotating wheel, and onto another mirror approximately 22 miles (35 km) away. The beam was reflected through the wheel and onto the first mirror. By the time the beam reached the rotating wheel again, the wheel had moved on to another slot. By measuring how fast the wheel was rotating, the spacing between the slots, and the distance from the more distant

mirror to the wheel, the two scientists could calculate the speed of light.

It was difficult to get an accurate figure for the speed of light using this method, so in 1850 Foucault replaced the spinning wheel with a rotating mirror. As the mirror swiveled, it reflected the returning beam to a slightly different position. This difference revealed the speed clearly. Foucault got a very **close calculation for the speed of light** at 185,168 miles per second (298,000 km per second).

Further advances in science include the publication in Germany of physicist **Rudolf Clausius's** (1822–88) 1850 paper on the movement of heat. He laid **foundations for the science of thermodynamics** with two basic laws. The first is conservation of energy—energy is never lost, but simply redistributed. The second is that heat can never move from a cold place to a hot one, only the reverse.

In 1850, British chemist **James Young** (1811–83) patented a method of **distilling paraffin** from coal, which gradually replaced whale oil in domestic



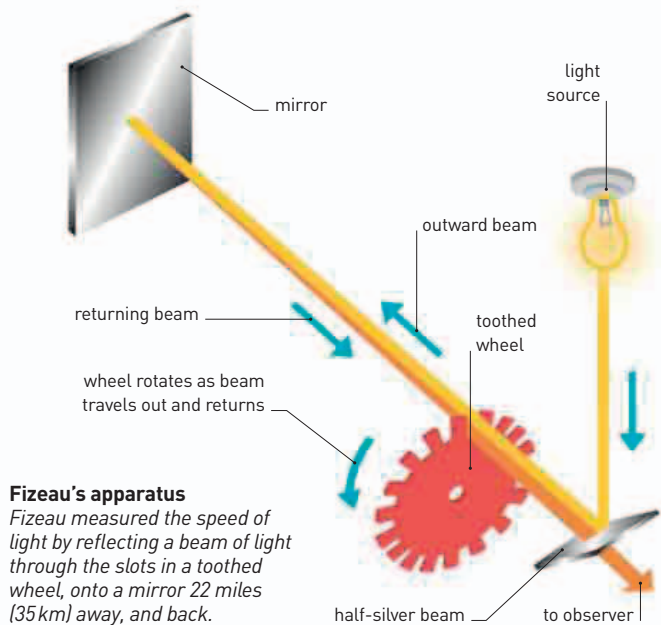
oil lamps. Scotch-born American inventor **John Gorrie** (1803–55) also **introduced refrigeration**, when he created a machine for making ice using circulating liquid to draw out heat (see 1872–73).

In 1851, British astronomer **William Lassell** (1799–1880) **discovered two more moons of Uranus**, Ariel and Umbriel, and British sculptor **Frederick Scott Archer** (1813–57) introduced the **wet plate photographic process**. This involved coating

Wet plate camera

Working in a portable darkroom, the photographer had under 10 minutes to take a picture and process it before the plate dried.

a photographic plate with a sticky liquid called collodion in darkness before each photo was taken. It had the fine detail of daguerreotypes (see 1837) and the repeatability of Fox Talbot's calotype (see 1835).



Fizeau's apparatus

Fizeau measured the speed of light by reflecting a beam of light through the slots in a toothed wheel, onto a mirror 22 miles (35 km) away, and back.

1849 Hippolyte Fizeau and Jean Foucault measure the speed of light

1850 Jean Foucault's improved apparatus gives an accurate figure for the speed of light

1849 Edouard Roche explains Saturn's rings by tidal forces

1850 Hippolyte Fizeau and E. Gounelle measure the speed of electricity

1850 James Young patents a method of distilling paraffin from coal

1851 Frederick Scott Archer introduces the wet plate photographic process

1850 Rudolf Clausius lays the foundations of thermodynamics with two basic laws

October 24, 1851 William Lassell discovers two more moons of Uranus